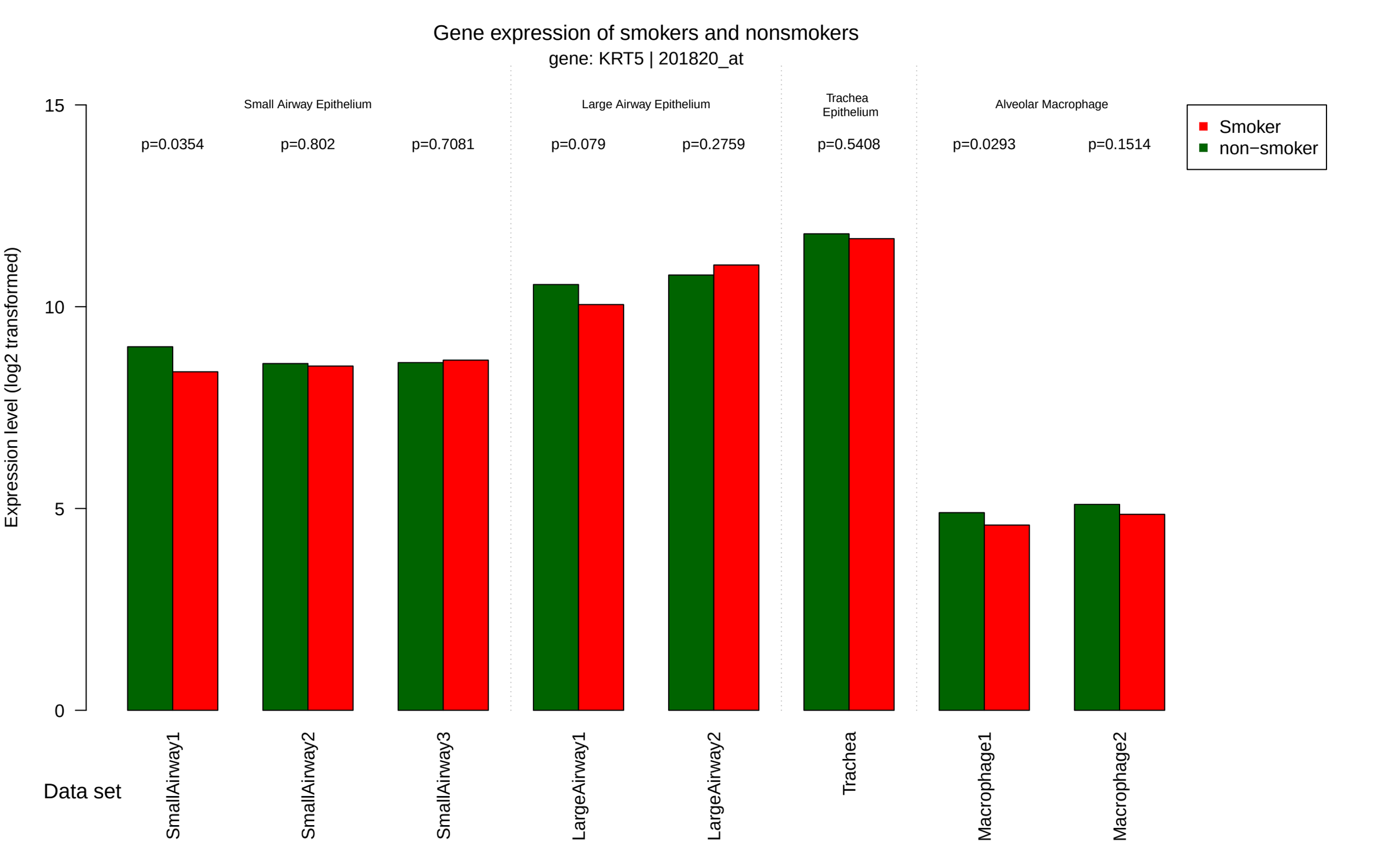
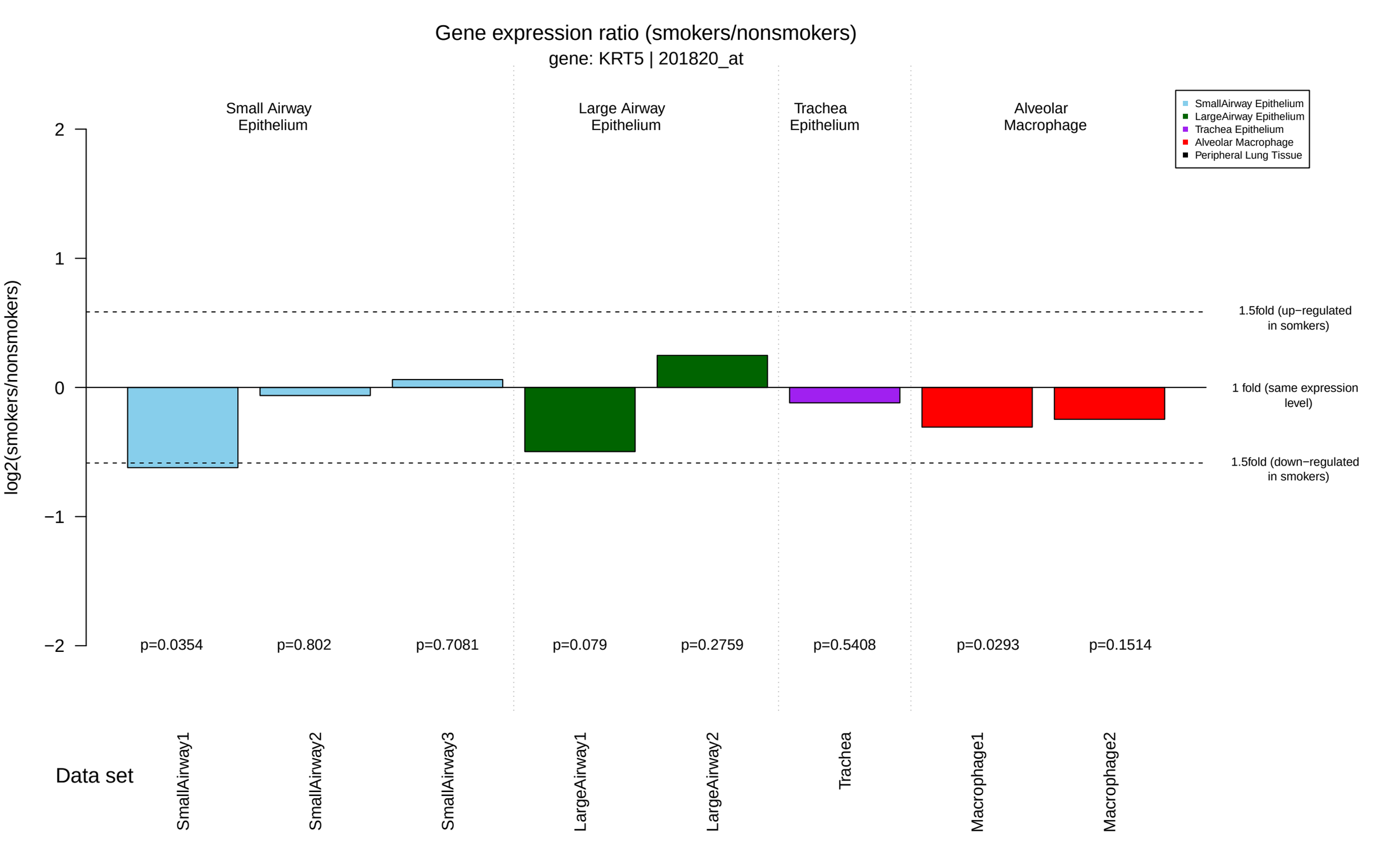
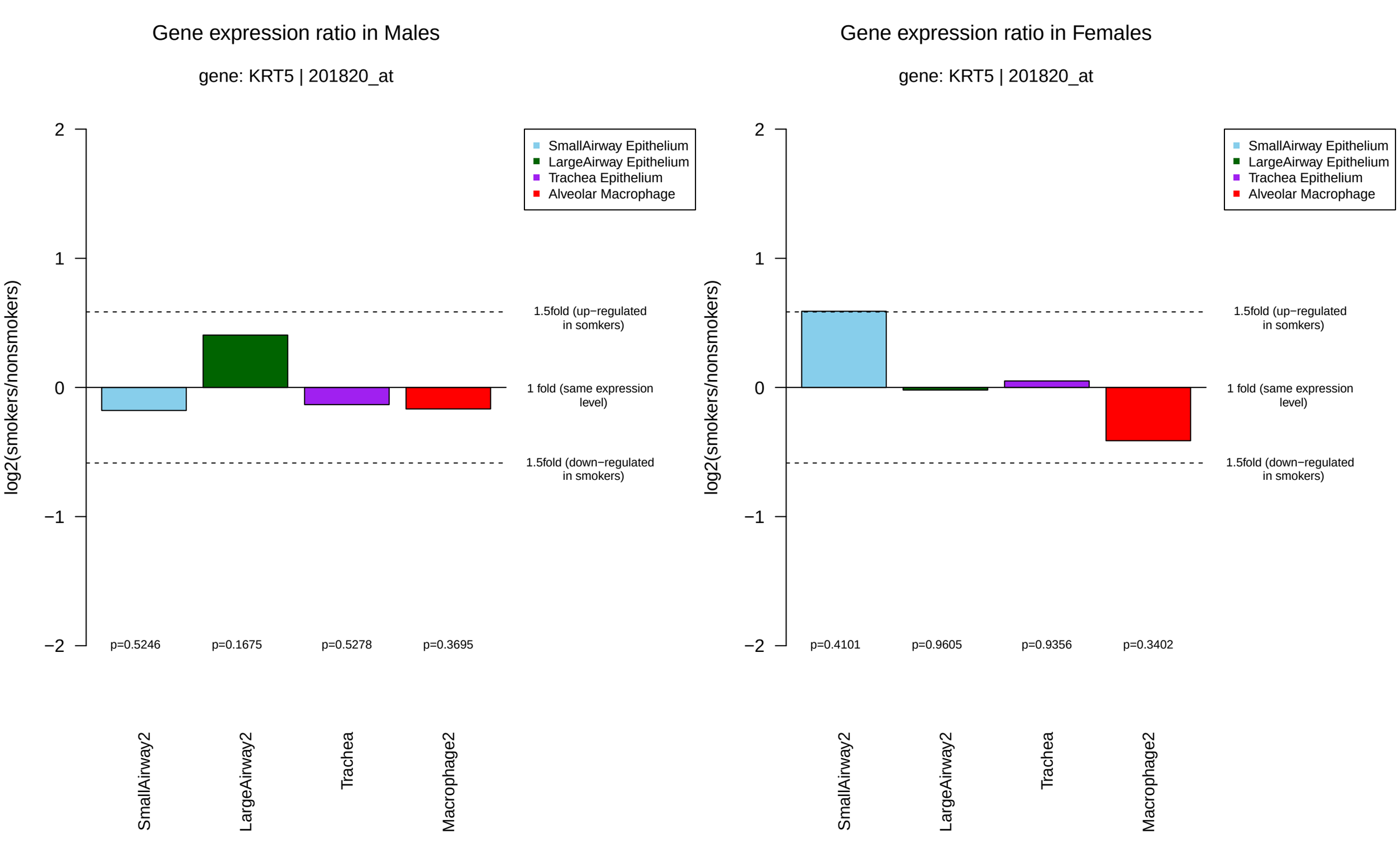
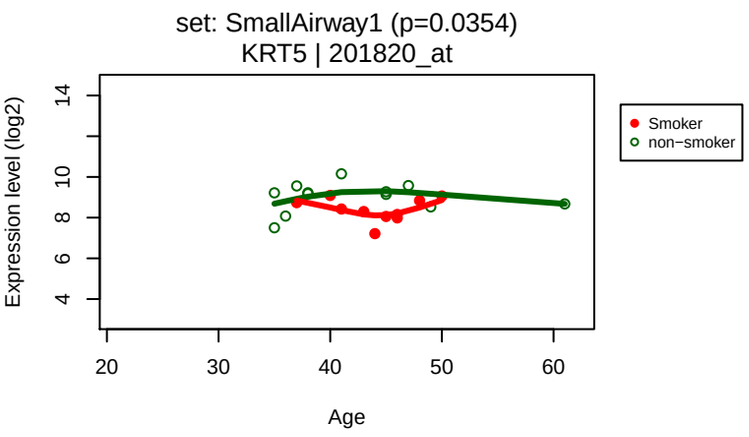
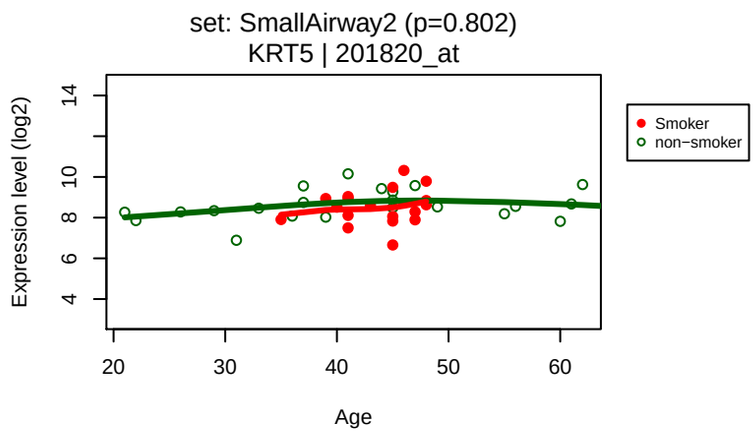
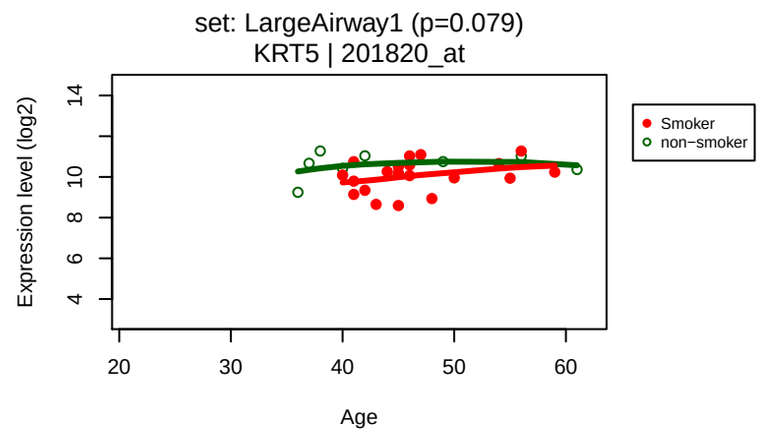

Gene expression of smokers and nonsmokers
gene: KRT5 | 201820_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0354
p=0.802
p=0.7081
p=0.079
p=0.2759
p=0.5408
p=0.0293
p=0.1514
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

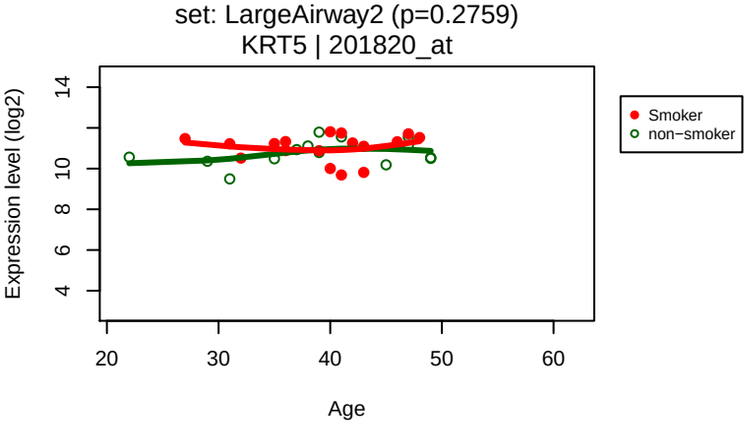

Gene expression ratio (smokers/nonsmokers)
gene: KRT5 | 201820_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0354
p=0.802
p=0.7081
p=0.079
p=0.2759
p=0.5408
p=0.0293
p=0.1514
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

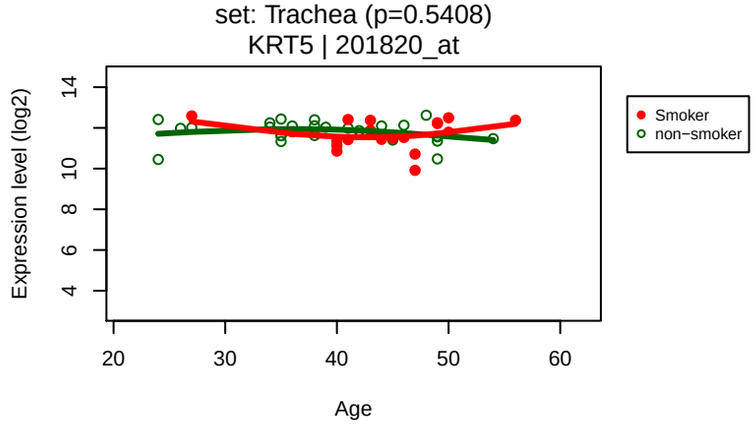

Gene expression ratio in Males
gene: KRT5 | 201820_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5246
p=0.1675
p=0.5278
p=0.3695
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KRT5 | 201820_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4101
p=0.9605
p=0.9356
p=0.3402
SmallAirway2
LargeAirway2
Trachea
Macrophage2

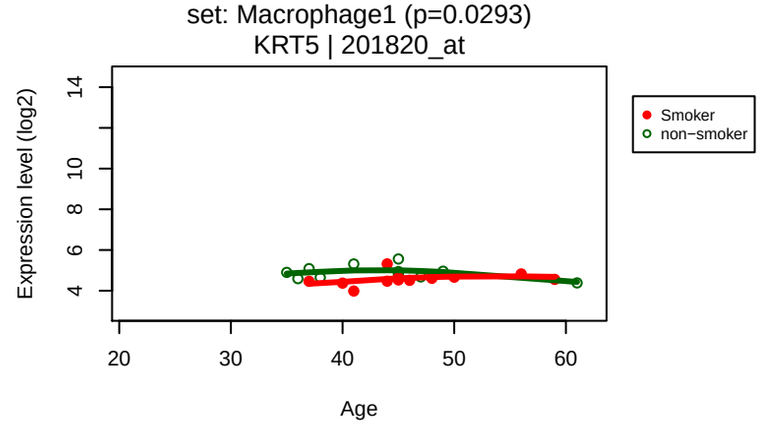

set: SmallAirway1 (p=0.0354)
KRT5 | 201820_at
Expression level (log2)
Age
Smoker
non−smoker

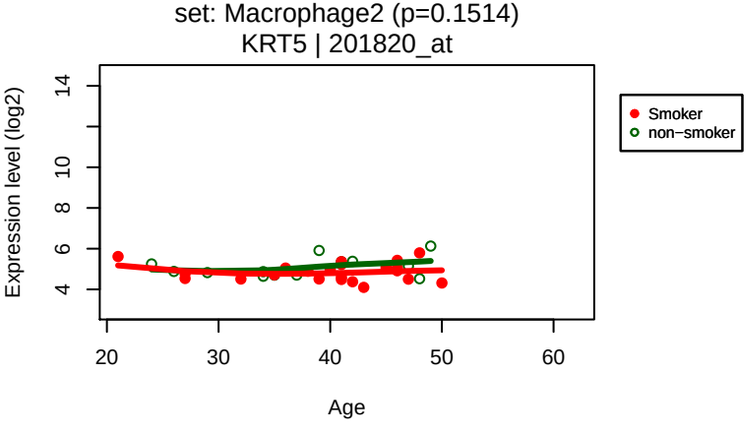

set: SmallAirway2 (p=0.802)
KRT5 | 201820_at
Expression level (log2)
Age
Smoker
non−smoker

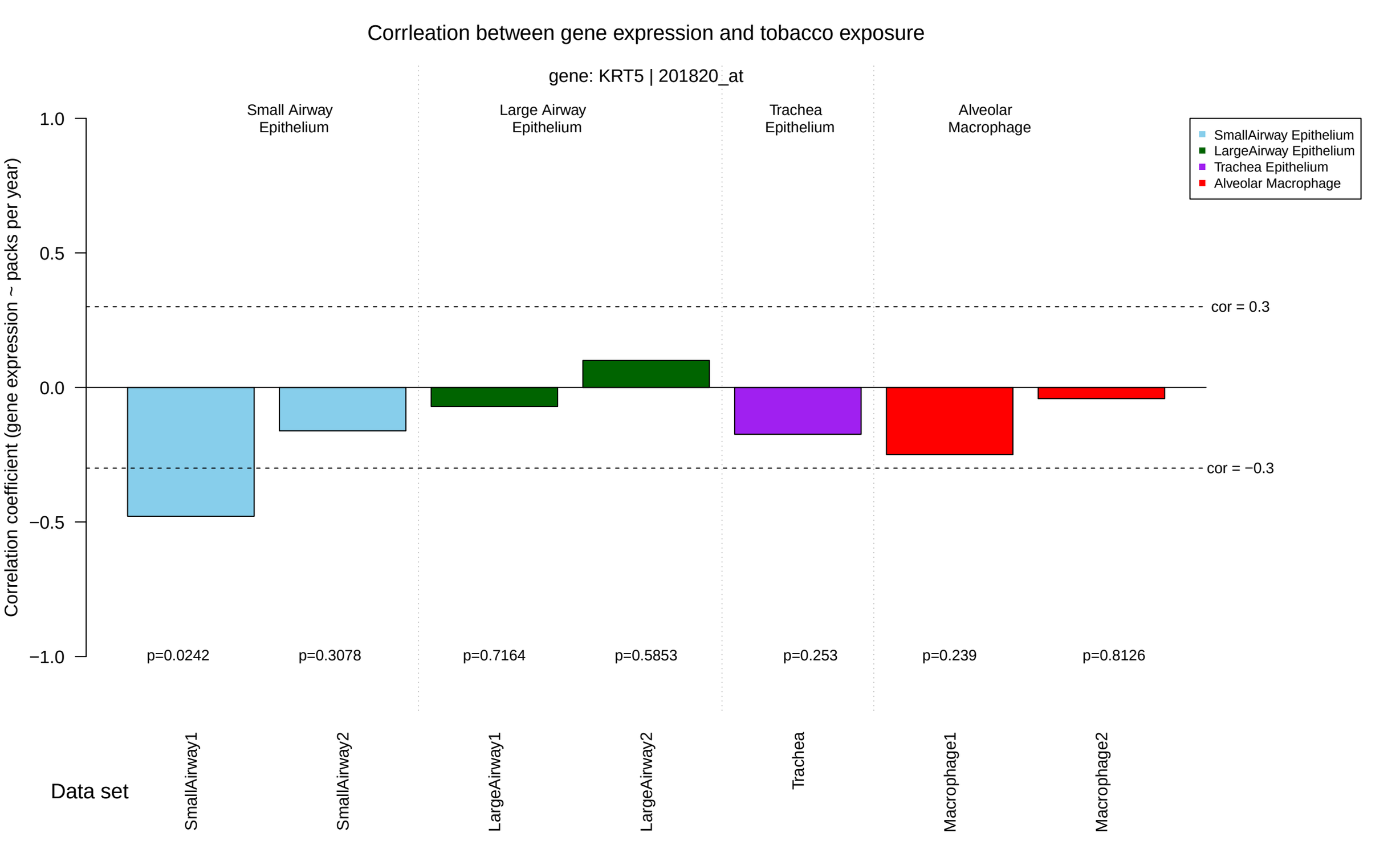

set: LargeAirway1 (p=0.079)
KRT5 | 201820_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2759)
KRT5 | 201820_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5408)
KRT5 | 201820_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0293)
KRT5 | 201820_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1514)
KRT5 | 201820_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KRT5 | 201820_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0242
p=0.3078
p=0.7164
p=0.5853
p=0.253
p=0.239
p=0.8126
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2